

Milestone 1: Data Collection and Preparation (Weeks 1–2)

1. Introduction

Milestone 1 represents the foundational phase of the Global Defense Metrics (GDM) project. The primary objective of this milestone is to build a reliable and reproducible data pipeline capable of collecting, structuring, and preparing large-scale global military data for analytical use. Since all subsequent analytics, KPIs, and dashboards depend directly on the quality of the underlying data, this milestone focuses entirely on **how the data was obtained, understood, and prepared**, rather than on evaluation or visualization outcomes.

The work carried out during this milestone spans the first two weeks of the project timeline and is divided into two modules:

- Module 1: Scraping Setup and Execution
- Module 2: Data Cleaning and Structuring

Together, these modules establish a clean, well-understood data foundation that supports flexible analytics and dashboard development in later stages of the project.

2. Module 1: Scraping Setup and Execution

2.1 Objective

The objective of Module 1 is to collect country-level military, manpower, economic, and infrastructure metrics for more than 140 countries from GlobalFirepower.com using Python. The focus of this module is not only on extracting data, but also on **understanding the structure of the source website** and designing a scraping approach that is accurate, scalable, and reproducible.

2.2 Exploration of Data Source

Before implementing the scraping logic, the GlobalFirepower website was explored manually to understand how data is organized and presented. The base URL and multiple metric-specific pages were examined to study their HTML structure. Using Python and BeautifulSoup, the page source of these metric pages was inspected to identify consistent patterns for:

- Country names
- Metric values
- Ranking and non-relevant visual elements

Through this exploration, it was observed that GlobalFirepower uses structured HTML blocks to represent metric records, rather than traditional HTML tables. Understanding this structure was critical in designing a scraper that extracts only meaningful data while ignoring visual or ranking-related artifacts.

2.3 Scraping Design and Implementation

Based on the insights gained from page source exploration, a metric-centric scraping strategy was implemented. Instead of scraping individual country pages, each metric page was processed independently, extracting values for all countries listed on that page in a single pass.

The scraping logic was implemented in Python using the following key steps:

- Reading metric URLs from the predefined file `links_for_military_data.txt`
- Sending HTTP requests to each metric page
- Parsing the HTML content using BeautifulSoup
- Extracting long-form country names and corresponding metric values
- Ignoring ranking numbers, short country codes, and visual-only elements

This approach ensured consistent country coverage across metrics and reduced redundancy in page requests.

2.4 Raw Data Generation

All successfully extracted metrics were consolidated into a single raw dataset named `military_raw_data.csv`. In this dataset:

- Each row represents one country
- Each column represents one scraped metric
- Values are stored in their original textual form as obtained from the website

No cleaning or transformation was applied at this stage. The purpose of this dataset is to act as a faithful, unaltered representation of the scraped data, allowing for transparency, debugging, and reprocessing if required.

The scraping logic was encapsulated in the Python script `scrape_military_metrics.py`, completing the data acquisition phase of the milestone.

3. Module 2: Data Cleaning and Structuring

3.1 Objective

Module 2 focuses on transforming the raw scraped dataset into a clean, structured, and analytics-ready format. While the raw dataset accurately reflects the source data, it is not immediately suitable for analysis due to inconsistent data types, formatting symbols, and the presence of metrics that are not analytically necessary.

The goal of this module is to:

- Select only relevant metrics
- Convert all quantitative values into numeric formats
- Handle missing values responsibly
- Produce a clean dataset that can be directly used in BI tools and analytical workflows

3.2 Metric Selection Strategy

During this stage, the raw dataset was examined to determine which metrics were actually required for meaningful military analysis. It was observed that not all scraped metrics contribute directly to comparative analysis, KPI derivation, or coalition-level aggregation.

Metrics were selected based on the following criteria:

- Relevance to military strength assessment
- Usefulness for comparative and per-capita analysis
- Applicability in coalition and aggregation scenarios
- Analytical value rather than visual redundancy

This selection process reduced noise in the dataset while preserving flexibility for advanced analytics. Only the most meaningful manpower, air, land, naval, economic, and infrastructure indicators were retained for further processing.

3.3 Data Type Normalization

After selecting the required metrics, the dataset was analyzed to inspect data types. It was observed that a majority of the metric columns were stored as objects due to the presence of commas, currency symbols, percentage signs, and other formatting characters.

A uniform cleaning procedure was applied to remove such symbols and normalize the values. Following this, all metric columns were converted into appropriate numeric data types using safe conversion techniques. This ensured consistency across the dataset and enabled mathematical operations, aggregation, and visualization.

3.4 Missing Value Identification and Handling

Once numeric conversion was completed, missing values were identified across the dataset. Instead of applying blanket imputation, missing values were handled using context-aware logic:

- For asset-based metrics, missing values were interpreted as zero where absence logically indicates non-availability of that asset
- For population and economic metrics, missing values were retained to avoid introducing artificial data

Through this approach, the dataset maintained analytical integrity while significantly reducing sparsity. After cleaning and imputation, the overall proportion of missing values in the dataset was reduced to below 2 percent.

3.5 Clean Dataset Generation

The final outcome of Module 2 is the cleaned and structured dataset `military_cleaned.csv`. This dataset:

- Contains only analytically relevant metrics
- Uses consistent numeric data types
- Has minimal missing values
- Is compatible with Tableau, Power BI, and Python-based dashboards

All cleaning and transformation steps were implemented and documented in the Python notebook `clean_data.ipynb`.

4. Milestone 1 Summary

Milestone 1 successfully establishes the data backbone for the Global Defense Metrics project. Through careful exploration of the data source, thoughtful scraping design, and systematic data cleaning, the project achieved a reliable and transparent data preparation pipeline.

The milestone concludes with two key datasets:

- A raw dataset that preserves the original scraped values
- A clean dataset optimized for analytics and visualization

These outputs provide a solid foundation for subsequent milestones, which focus on KPI derivation, advanced comparative analysis, and interactive dashboard development.